

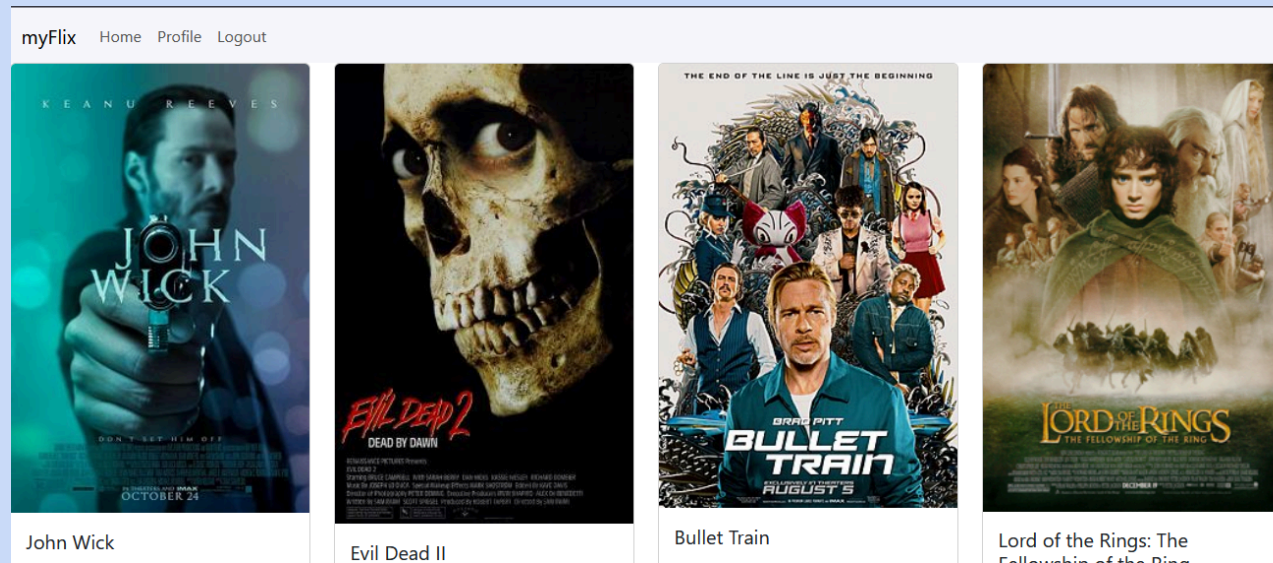
Case Study of myFlix Full-Stack Project

Overview

myFlix is a project developed using the MERN stack, entirely by one developer. This project exists to provide users with information about movies, including

- Directors,
- Genres,
- and Descriptions.

Users can create an account which saves their login and allows them to save movies to a list of favorites.



A picture of the Javascript code for myFlix's API.

```
/**
 * The welcome page
 * @returns a welcome page
 */
app.get('/', (req, res) => {
  res.send('Welcome to my movie club!');
});

/**
 * This endpoint gives a list of all movies
 * @returns a JSON of the movies list within the database
 */
app.get(
  '/movies',
  passport.authenticate('jwt', { session: false }),
  async (req, res) => {
    await Movies.find()
      .then((movies) => {
        res.status(201).json(movies);
      })
      .catch((err) => {
        console.error(err);
        res.status(500).send('Error: ' + err);
      });
  }
);
```

A picture of the main page of myFlix.

Purpose and Context

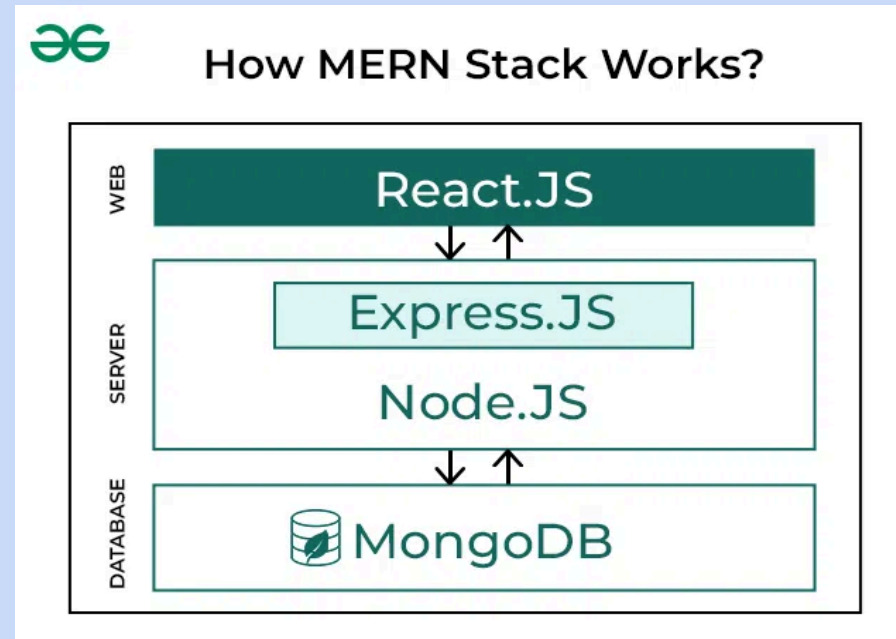
myFlix was created as a part of a Full-Stack Immersion course with CareerFoundry, as a personal project meant to introduce its developer to MongoDB, Express, React, and Node.js. myFlix exists as the frontend application of an API project this developer had already created.

Objective

This project was a great way to introduce oneself to the MERN stack, and makes a good addition to a portfolio. myFlix exists as proof of a developer's learning journey.

Duration

myFlix's development was made of up 20 deliverables, more than double most other projects in this developer's portfolio. This, combined with the adjustment period for learning React, did cause delays in delivering the final project.



"How does MERN stack work?" A graphic from GeeksforGeeks explaining the MERN stack.

Credits

Lead Developer: Alexander Eggleston

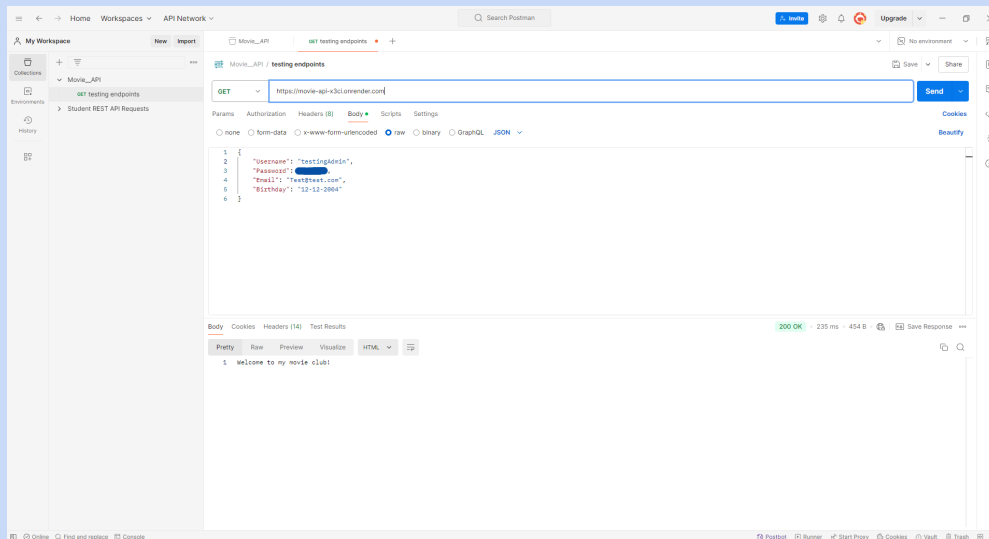
Tutor: Ezequiel De Simone

Mentor: Enrique Gonzalez



Server-Side Development

The server and database development came first, as otherwise no information would show up on the frontend. The API follows RESTful endpoint principles, and uses



A picture of Postman, the API testing app, testing myFlix's endpoints.

Postman was used to test the API, as well as its verification system using JWT tokens. Tokens are like cookies that a site will give a browser to remember its login information. Specifically, the API will give out tokens that last seven days.

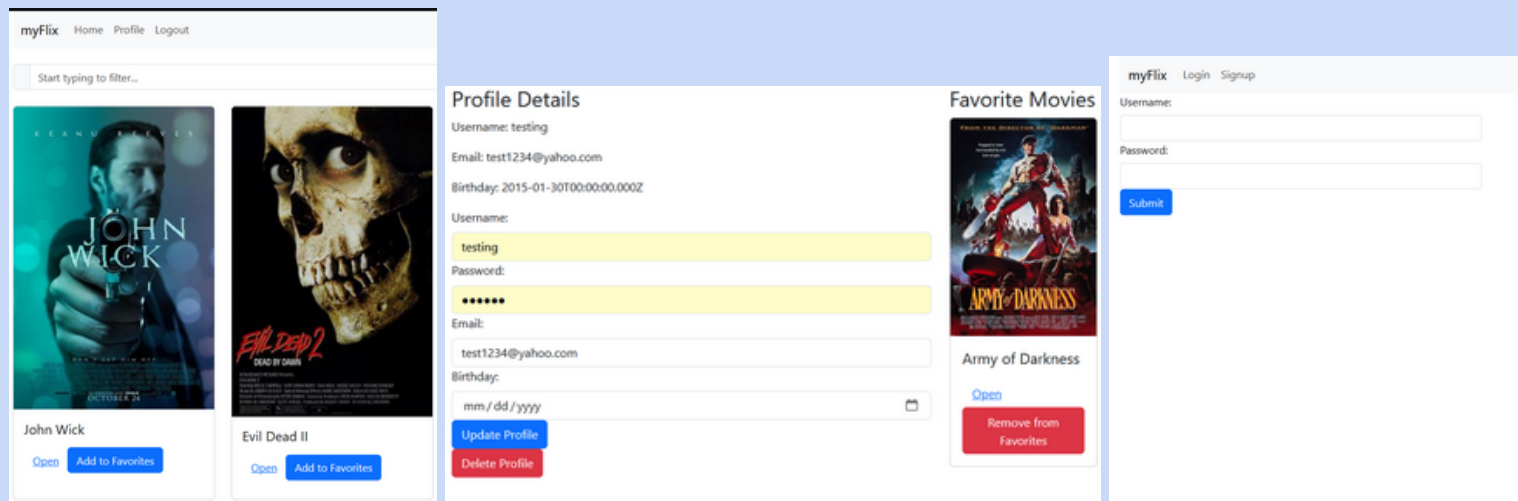
Client-Side Development

The myFlix client is a single page, responsive application developed with React and React Redux. It is what users interact with to send and receive information to the database using the API. React uses a “View” structure, so the pages are broken down as,

- Main View: What a user will see upon opening the application
- Login View: How a user either creates a user or logs in
- Movie View: Specific information about a movie
- User View: How a user finds their own information and their favorites list

Express and Node.js. It interacts with a non-SQL database through MongoDB, and one may note this is already three letters of the MERN stack.

The API uses HTTP methods such as GET and POST to perform CRUD operations in the database. CRUD stands for Create, Read, Update, and Destroy, it refers to methods used to interact with information in databases. The API's main function is to return information about movies as JSON objects for the user.



In order: Movie View, User View, and Login View

Challenges

The greatest challenge the developer faced was the implementation of the favorites list. It started as a simple idea, adding a movie's internal ID to a list within each existing user, but the challenge arose in trying to add a movie to favorites on the Movie View, and saving it to the User View. For whatever reason, the “Add to Favorites” button was clearly not updating the database. The reason the database was not updating was frustratingly simple. The developer named the list within each user “Favorites”, but was trying to add movie ID's to a list called “Favorite_Movies”.





Reflection

myFlix is still this developer's favorite project, as it introduced him to many important languages, concepts, and techniques. It taught him about the importance of keeping variable names clear, as well as the difficulties inherent to uploading a project online. In a way, it was his first **real** project, a start-to-finish website that works all the way through, and the only way to improve it would be to add every movie to the database. This developer wishes every project was more like myFlix.